

Few-Shot Personalization for Consistent Character Generation

Tongjia Rao Haoshen Wu Jiasheng Zhu
{tr426, hw862, jz2455}@cornell.edu

1. Problem Statement

We study **few-shot personalization**: how many reference images are needed for consistent identity, and how the personalization method changes the trade-off between fidelity and efficiency. We compare Textual Inversion and DreamBooth+LoRA under 3, 5, 10, and 20 training images. DreamBooth+LoRA follows standard parameter-efficient fine-tuning, reducing memory and storage relative to full DreamBooth while keeping strong identity fidelity. The main challenge is balancing identity consistency, prompt control, and training cost under scarce data.

2. Related Work

LoRA [1] enables efficient fine-tuning by adding small trainable low-rank matrices to existing layers. In DreamBooth, it cuts trainable parameters and checkpoint size while preserving much of the personalization quality, making DreamBooth+LoRA a practical standard for subject-driven generation. Custom Diffusion [2] further shows that tuning only selected diffusion layers can be effective, reinforcing our focus on parameter-efficient personalization. For evaluation, we use DINOv2 [3] for identity similarity and CLIP [4] for text-image alignment. Prior work shows strong qualitative results, but controlled low-shot comparisons remain limited; our project benchmarks minimum useful dataset size across methods.

3. Methodology

- **Dataset:** We will choose one target subject (e.g., a pet, plushie, or object) and collect 20–30 curated photos. From this pool, we will sample subsets of size 3, 5, 10, and 20, with 3 random trials per size. Images will be cleaned to 1024×1024 and checked for diversity in viewpoint, background, and lighting.
- **Architecture/Algorithms.**
 - **Method A – Textual Inversion:** Learn a new token embedding (e.g., <my-character>) for the target while keeping model weights frozen.
 - **Method B – DreamBooth+LoRA:** Fine-tune the model with a rare identifier token and prior-preservation loss, using LoRA adapters (rank 16–64) in the U-Net cross-attention layers.
 - **Base model:** Stable Diffusion XL (stabilityai/stable-diffusion-xl-base-1.0).
- **Evaluation.** We will use a fixed prompt suite spanning style, scene, pose/action, and accessories, with fixed random seeds for reproducibility. Metrics:

- **Identity consistency:** cosine similarity of DINOv2 features between reference images and generated images.
- **Prompt adherence:** CLIP text-image alignment score.
- **Image quality:** human ranking and artifact counts.
- **Efficiency:** training time and peak GPU memory per method.

4. Deliverables and Expected Results

- Reproducible training and evaluation pipeline for both methods.
- Full experiment matrix: 2 methods $\times$ 4 shot counts $\times$ 3 trials = 24 runs.
- Quantitative tables and plots of identity score vs. training-set size, with variance across trials.
- Qualitative comparison grids and failure-case analysis.
- Final report and presentation.

Expected results: We expect DreamBooth+LoRA to produce better identity fidelity in low-shot settings, while Textual Inversion should train faster and use less memory. We also expect diminishing returns beyond roughly 10–15 images. The final outcome will be practical guidance on the minimum dataset size needed for each method under limited compute.

5. Feasibility

This project is feasible within one semester because it relies on pretrained models and established open-source implementations rather than new architectures or training from scratch. We will use Hugging Face Diffusers and run a structured experiment grid. Each Textual Inversion run should take about 1–2 hours, and each DreamBooth+LoRA run about 30 minutes to 1 hour on a single A100 GPU. The full 24-run matrix is therefore about 40–70 GPU-hours, which is manageable within a semester. SDXL needs roughly 16–24 GB VRAM, within a single A100, and the modular design means partial results would still be informative if some runs fail.

6. Workload Distribution

Student 1 (Tongjia Rao): experiment design, evaluation framework (DINO/CLIP), and fairness protocol.

Student 2 (Haoshen Wu): Textual Inversion pipeline, hyperparameter sweeps, and qualitative result curation.

Student 3 (Jiasheng Zhu): DreamBooth+LoRA pipeline, LoRA and prior-preservation tuning, and compute/logging management.

Shared: dataset collection, analysis writing, figure polishing, and presentation preparation.

References

- [1] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhen Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *ICLR*, 2022.

- [2] Nupur Kumari, Bingliang Zhang, Richard Zhang, Eli Shechtman, and Jun-Yan Zhu. Multi-concept customization of text-to-image diffusion. *CVPR*, 2023.
- [3] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.
- [4] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. *ICML*, 2021.